Sheet 7 - Search

Q1. $\boxed{T=2}$ $u_t = \sqrt{c_t}$. Let remaining cake at each point be x_t.

The problem is $\max_{c_0, c_1, c_2} \sqrt{c_0} + \delta\sqrt{c_1} + \delta^2\sqrt{c_2}$

In period 2, $c_2 = x_2$, $V_2(x_2) = \sqrt{x_2}$.

In period 1, $c_1 = x_1 - x_2$, $V_1(x_1) = \sqrt{x_1 - x_2} + \delta\sqrt{x_2}$

So optimising wrt c_1, which is a function of x_2,

$$\frac{\partial V(x_1)}{\partial x_2} = -\frac{1}{2}(x_1 - x_2)^{-1/2} + \frac{1}{2}\delta(x_2)^{-1/2} \overset{set}{=} 0$$

$$\Rightarrow x_1 - x_2 = \delta^{-2} \cdot x_2 \qquad = x_1 - \frac{x_1}{1+\delta^{-2}}$$

$$x_1 = (1+\delta^{-2})\, x_2.$$

In period 0, $c_0 = x_0 - x_1$

and $V(x_0) = \sqrt{x_0 - x_1} + \cancel{\delta\sqrt{\delta^{-2}x_2}} + \delta^2\sqrt{\frac{1+\delta^{-2}}{x_1}}$

$$\delta\sqrt{x_1\left(1 - \frac{1}{1+\delta^{-2}}\right)}$$

$$\frac{1}{1+\delta^{-2}}$$

Now optimising wrt c_0, a function of x_1

$$\frac{\partial V(x_0)}{\partial x_1} = -\frac{1}{2}(x_0 - x_1)^{-1/2} + \frac{1}{2}\delta\sqrt{1 - \frac{1}{1+\delta^{-2}}} \cdot x_1^{-1/2} + \frac{1}{2}\delta^2 \cdot \left(\frac{x_1}{1+\delta^{-2}}\right)^{-1/2}$$

$$\left(1 - \frac{1}{1+\delta^{-2}} = \frac{\delta^{-2}}{1+\delta^{-2}} = \frac{1}{1+\delta^2}\right)$$

$$\Rightarrow (x_0 - x_1)^{-1/2} = \frac{\delta}{\sqrt{1+\delta^2}}\, x_1^{-1/2} + \delta^3 \cdot \frac{1}{\sqrt{1+\delta^2}}\, x_1^{-1/2}$$

since $\delta^2 \cdot (1+\delta^{-2})^{1/2} \cdot (1+\delta^{-2})^{-1} = \delta^2 \cdot (1+\delta^{-2})^{-1/2}$

$$= \delta^2 \cdot \delta/\sqrt{1+\delta^2} = \delta^3/\sqrt{1+\delta^2}$$

So $(x_0 - x_1)^{-1/2} = x_1^{-1/2}\left(\frac{\delta(\delta^2+1)}{\sqrt{\delta^2+1}}\right) = x_1^{-1/2}\,\delta\sqrt{\delta^2+1}$

$$\therefore \frac{1}{x_0 - x_1} = \frac{1}{x_1} \cdot \delta^2(\delta^2+1)$$

$$(1 + \delta^2(\delta^2+1))\, x_1 = x_0\,(\delta^2(\delta^2+1)) \Rightarrow x_1 = \frac{\delta^4 + \delta^2}{1+\delta^2+\delta^4}\, x_0$$

So the optimal strategy, you should consume (with $x_0 = 1$)

$$c_0 = x_0 - x_1 = \frac{1}{1+\delta^2+\delta^4}$$

$$c_1 = x_1 - x_2 = \frac{\delta^2}{1+\delta^2+\delta^4}$$

Q1 all correct!

$$c_2 = x_2 = \frac{\delta^4}{1+\delta^2+\delta^4}$$

$\boxed{T=\infty}$ a) $V(x_t) = \max\limits_{x_{t+1}\in\Gamma(x_t)} \left\{\sqrt{x_t - x_{t+1}} + \delta V(x_{t+1})\right\}$, where $\Gamma(x_t) = [0, x_t]$

since you can't carry forward more cake than you have, nor a negative amount.

Q2a-b: correct

b) We had $V_2(x_2) = \sqrt{x_2}$, $V_1(x_1) = \sqrt{1+\delta^2}\sqrt{x_1}$.

So likely form of value function is $V(x_t) = C\sqrt{x_t}$

c) $C\sqrt{x_t} = \max\limits_{x_{t+1}} \left\{\sqrt{x_t - x_{t+1}} + \delta C\sqrt{x_{t+1}}\right\}$, substituting in to BE

Taking FOC for RHS, $\frac{\partial}{\partial x_{t+1}} = -\frac{1}{2}(x_t - x_{t+1})^{-1/2} + \frac{1}{2}\delta C (x_{t+1})^{-1/2}$

$$\Rightarrow \delta^{-2}C^{-2}x_{t+1} = x_t - x_{t+1}, \quad x_{t+1}^* = \frac{x_t \cdot \delta^2 C^2}{1+\delta^2C^2}$$

$$x_t - x_{t+1}^* = x_t\left(\frac{1}{1+\delta^2C^2}\right) = x_t \cdot \frac{\delta^2C^2}{1+\delta^2C^2}$$

$$\therefore C\sqrt{x_t} = \sqrt{x_t \cdot \frac{\delta^2C^2}{1+\delta^2C^2}} + \delta C\sqrt{\frac{x_t}{1+\delta^2C^2}}$$

$$= \delta C \cdot \sqrt{x_t} \cdot \left(\frac{1}{\sqrt{1+\delta^2C^2}}\right)$$

[working slightly wrong]

So $C = \sqrt{1+\delta^2C^2}$, $C^2 = \frac{1}{1-\delta^2}$

$$\therefore C = \frac{1}{\sqrt{1-\delta^2}}$$

and thus the optimal policy is $x_{t+1} = \frac{\delta^2/(1-\delta^2)}{1+\delta^2/(1-\delta^2)} x_t = \delta^2 x_t$

Q2c yes, correct

So at each period consume $c_t = x_{t+1} - x_t = (1-\delta^2) x_t$, a fixed fraction of what remains.

d) We're interested in $V(x_0)$ for $x_0 = 1$, which is $\frac{1}{\sqrt{1-\delta^2}} \cdot \sqrt{1} = (1-\delta^2)^{-1/2}$.

Q2d correct!

Greater impatience (smaller δ) makes the denominator $\sqrt{1-\delta^2}$ larger since $1-\delta^2 \in [0,1]$, so your lifetime utility is lower.

As $\delta \to 1$, $V(1) \to \infty$ and as $\delta \to 0$, $V(1) \to 1$, since you eat it all at $T=0$.

2. $u \sim U(0,1)$; $F(x) = x$ for $x \in [0,1]$.

a) After opening n boxes, my ~~expected~~ best prize's distribution is found by $\hat{u} = \max\{u_1, \dots, u_n\}$

$P(\hat{u} \le x) = \prod_{i=1}^{n} P(u_i \le x) = x^n$. So $\mathbb{E}$ prize is $n\int_0^1 x \cdot x^{n-1} dx = n\int_0^1 x^n dx$

The expected gains from searching a marginal n^{th} box, ~~being searched~~, are

$$g(n) = (n+1)\int_0^1 x^{n} dx - (n)\int_0^1 x^{n-1} dx$$

$$= \frac{n}{n+1} - \frac{n-1}{n} = \frac{n^2 - (n+1)(n-1)}{n(n+1)} = \frac{1}{n(n+1)}.$$

So ~~$g(0) = 2/\ldots$~~, $g(1) = \frac{1}{2}$, $g(2) = \frac{1}{6}$, $g(3) = \frac{1}{12}$.

Thus optimally:

- $s < 1/12 \Rightarrow$ search all 3
- $1/12 \le s < 1/6 \Rightarrow$ search 2
- $1/6 \le s < 1/2 \Rightarrow$ search 1.

b) We can use backward induction. At $T=3$ you must take whatever is in the final opened box, so reservation value $z_3 = 0$.

At $T=2$, z_2 is pinned down by indifference between taking an existing prize of value z_2, and continuing (which means you end up with x_3 for sure and pay s more).

So $z_2 = \mathbb{E}(x_3) - s = \frac{1}{2} - s$.

At $T=1$, continuing is more complicated – you'd optimally get x_2 iff $x_2 \geq z_2$, and x_3 otherwise.

$$\text{So} \quad z_1 = -s + \underbrace{P(x_2 \geq z_2)}_{(1-P(x_2 \leq z_2))} \mathbb{E}[x_2 \mid x_2 \geq z_2] + P(x_2 < z_2) \cdot (\mathbb{E}x_3 - s)$$

$$= -s + (1-z_2)\left(\frac{z_2+1}{2}\right) + z_2 \cdot \left(\frac{1}{2} - s\right)$$

$$= -s + \underbrace{\frac{z_2}{2}}_{\frac{1}{4}-s/2} + \frac{1+z_2^2}{2} - \underbrace{z_2 s}_{\frac{s}{2}-s^2}$$

$$= -s + \frac{1}{4} - \frac{s}{2} + \frac{1}{2} + \frac{(\frac{1}{2}-s)^2}{2} - \frac{s}{2} + s^2$$

Which could be simplified. And then the optimal strategy is to keep prize u_t if it's $\geq z_t$, else proceed to next box. Larger values of s mean you're less picky, i.e. willing to take a lower prize sooner, since search gains are less worth while.

c) Free recall gives stationary reservation price, because you're never at risk of "running out" of options.

We can pin down z by $H(z) = s$, since at any point the net benefit from opening another box is $H(\hat{x}) - s$ where $\hat{x}$ is the best prize so far, and $H(x)$ is expected amount by which next box exceeds x.

$$\text{Here} \quad H(x) = \int_x^1 (u-x)\,du = \frac{(1-x)^2}{2}$$

$$\Rightarrow \quad z = 1 - \sqrt{2s} \quad \text{which is} > 0 \text{ given } s < 1/2.$$

d) In c), P(stopping) is $P(x_t > z) = 1 - P(x_t \leq z)$ for each step t, which is $\sqrt{2s}$. ~~So expected number of searches~~

The expected number of boxes to open is
$$1 + (1 - \sqrt{2s}) + (1-\sqrt{2s})^2 \text{ which is } > 1 \text{ and } < 3.$$
So it's only the middle scenario where fixed sample opens 2 that we might have $\mathbb{E}$ boxes opened crossing over.

4. a) You would always open the first box, which contains prize u_1.

Then, when considering the second box, the expected net value would be
$\mathbb{E}[\max\{u_1 - b, u_2\}] - s$.

The reservation value z makes you indifferent between stopping and searching, so
$$z = \mathbb{E}[\max\{z-b, u\}] - s.$$
$$= -s + (z-b) + \underbrace{\int_{z-b}^{\bar{u}=1} (u-(z-b))\, dF(u)}_{H(z-b)}$$

So $H(z-b) = s + b$.

Thus you should open box 2 iff $u_1 < z$ pinned down above by H, which is the expected gross gains from search function and depends on F,
$$H(x) :\!= \mathbb{E}[\max\{u - x, 0\}]$$

b) H is a decreasing, convex, strictly positive function that $\to 0$ as $x \to 1$ and $\to \mathbb{E}[u]$ as $x \to 0$.

- With larger s, z must decrease (you get less picky if search costlier).
- With larger b, we need to use IFT (assumptions hold)
$$\frac{dz}{db} = -\frac{\partial G/\partial b}{\partial G/\partial z} \quad \text{where } G: H(z-b) - s - b \equiv 0$$
$$= -\frac{-H'(z-b) - 1}{H'(z-b)} = 1 + \frac{1}{H'(z-b)}$$ ~~which~~ is

with $H'(z-b) = -(1 - F(z-b)) \in [-1, 0)$, so $\frac{dz}{db} < 0$
hence z is decreasing in b. (Costlier recall makes search more expensive)

~~c) Return would never happen on optimal path if either~~

~~i) You only search the first box, so no return to do~~

~~ii) $u_1 - b < u_2$ for all u_1 and u_2, so return never worth it.~~

~~In case ii), this requires $b > u_1 - u_2$ for largest u_1 and smallest u_2, i.e. $b > 1$.~~

c) On the optimal path, Pandora only opens box 2 if $u_1 < z$.
And return would never happen iff for the best possible u_1 and worst u_2, we still have $u_2 > u_1 - b$.
So let $u_1 = z$ at the threshold to search, then with $u_2 = 0$, the condition is that $b > z$ to never return.

Alternatively, she might never return if she never opens box 2 (since no return to do). But for $z > 0$, search of box 2 happens with positive probability $F(z)$. ~~And~~ And we do have $z > 0$ because $H(0-b) = s+b$ isn't a solution: ~~as~~

$$H(0-b) = \mathbb{E}[u] + b = \tfrac{1}{2} + b$$

but $s < 1/2$, $\neq 1/2$ and solution needs larger z, for smaller LHS, as decreasing H.

We need to find z.

$$\cancel{H(z-b) = \int_{z-b}^{1} (u-(z-b))\,du}$$

At the threshold where return never happens $b = z$ and so

$$H(z-b) = H(0) = s + b = s + z \quad \text{But} \quad H(0) = \mathbb{E}[u] = \tfrac{1}{2} = s + z$$

So if b is larger than $z = 1/2 - s$, we don't have return, i.e. $b > 1/2 - s$.

d) Second box is never opened iff $u_1 \geq z$ w.p. 1, i.e. $z \leq 0$.
And since z is decreasing in s, the critical point is when $z = 0$.
So we have $H(-b) = \bar{s} + b$

$$\mathbb{E}[u] + b = \bar{s} + b \qquad \text{for } b \geq 0$$

$$\bar{s} = \mathbb{E}[u]$$ which doesn't depend on b; all that matters is the expected prize

e) (i) $(s,b) = (x,0)$: Then $H(v_1) = s = x$. $\Rightarrow v_1 = v_2 - x$

(ii) $(s,b) = (0,x)$. Then $H(v_2 - x) = b = x$.

Recall H is strictly decreasing in its argument. So given both have same RHS, but (ii) subtracts x from the reservation value, their arguments must $=$.

We have $v_1 < v_2$, so there's more search in the second case – it's free, so she's pickier about the first box's value to be willing to keep it without more search. To return to box 2 and pay b, that's a subset of the search box 2 worlds – so making search free is more important to reducing effective cost of opening box 2, as you only sometimes want to return but always pay s.

5. a) $\mathbb{E}[u_n] = \int_0^\infty u \cdot f_n(u)\, du$, where

$$f_n(u) = \frac{d}{du} F_n(u) = \frac{1}{n} e^{-u/n}$$

This is an exponential distribution with rate $1/n$, so $\mathbb{E}[u] = n$.

b) We have expected net ~~benefit~~ payoff from opening box n as

$$z_n = \mathbb{E}[\max\{z_n, u_n\}] - s_n$$
$$= z_n + \mathbb{E}[\max\{0, u_n - z_n\}] - s_n$$

so

$$s_n = H_n(z_n)$$
$$= \int_{z_n}^\infty (u_n - z_n) \cdot \frac{1}{n} e^{-u/n}\, du, \quad \text{and let } t_n = u_n - z_n, \; du = dt$$

$$= \int_{z_n}^\infty t \cdot \frac{1}{n} e^{-(t_n + z_n)/n}\, dt$$

$$= e^{-z_n/n} \int t \cdot \frac{1}{n} e^{-t_n/n}\, dt = e^{-z_n/n} \cdot \mathbb{E}[u] = n \cdot e^{-z_n/n}.$$

So $z_n = -n \ln \frac{s_n}{n} = n \ln(n/s_n)$

c) Pandora's rule is:

- If a box is to be opened, open the one with highest reservation value, i.e. the largest z_n.
- Stop opening boxes once the best available prize exceeds reservation values of all unopened boxes, including outside option 0.

d) $z_n = h_n \ln(u_n / s_n)$. So here $z_n = n \cdot \ln \frac{1}{a}$, where $u_n / s_n = \frac{1}{a}$ ($s_n = a n$)

$\frac{dz_n}{dn} = \ln \frac{1}{a}$ which > 0 iff $a < 1$, i.e. the highest reservation box in that case is $n = 10{,}000$.

She should open boxes working downwards from there, stopping as soon as she gets a prize larger than $n \cdot \ln \frac{1}{a}$ for the largest remaining n.

If $a > 1$, she ~~works~~ ~~opposed~~ has $-$ve reservation z_n so should take outside 0. If $a = 1$, she can sample randomly, or simply take 0.

e) We have $a < 1$, so Pandora does search.

$z_n = n \cdot \ln e = n$. So we want to find the box n s.t. $u_n = z_n$, i.e. $n^2 = 10000 \Rightarrow n = 100$, since for all earlier boxes $10{,}000/n < n$.